Al-Aqsa University
Faculty of Computers and IT
Dept. of Info & Computer science

جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

(ITCS4315)

Data Science

Assignment 3: Machine Learning

Presented to: **Eng. Nour Abo Rokba**

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Semester: First Semester 2025/2026

Assignment 3 – Submission

1. Products Dataset (Web Scraping – Assignment 1):

For the new dataset provided by the instructor (aistudentimpact_dataset.csv), I performed preprocessing by cleaning numeric columns, handling missing values, and removing duplicates. Then I applied two machine learning models. First, I used Decision Tree classification to predict Burnout Risk Level. The model achieved an accuracy of about 51%, with moderate performance across High, Medium, and Low categories, as shown in the confusion matrix. Second, I used Linear Regression to predict Post Semester GPA. The regression model achieved a strong R^2 score of 0.87 and a low Mean Squared Error of 0.03, indicating that the predictions were close to the actual GPA values. The scatter plot confirmed the accuracy of the regression results. Overall, classification gave moderate results, while regression performed strongly.

Google Colab link:
[https://colab.research.google.com/drive/1zgze0DXR7Xz8sbcttr3dpuAjwPd8BMuD?usp=sharing]

2. New Dataset (AI Student Impact Dataset):

For the new dataset provided by the instructor (ai_student_impact_dataset.csv), I performed preprocessing by cleaning numeric columns, handling missing values, and removing duplicates. Then I applied two machine learning models. First, I used Decision Tree classification to predict Burnout Risk Level. The model achieved an accuracy of about 51%, with moderate performance across High, Medium, and Low categories, as shown in the confusion matrix. Second, I used Linear Regression to predict Post Semester GPA. The regression model achieved a strong R^2 score of 0.87 and a low Mean Squared Error of 0.03, indicating that the predictions were close to the actual GPA values. The scatter plot confirmed the accuracy of the regression results. Overall, classification gave moderate results, while regression performed strongly.

Google Colab link:
[https://colab.research.google.com/drive/1SusH8WTrHOs6Xzh43tSslMLuyrrNoH_m?usp=sharing]

3. Video Explanation I recorded a video explaining both datasets, the machine learning steps, and the final results.

YouTube link: [https://youtu.be/mKd6yoxJqiw?si=XZszp2brb6McKkHb]

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